

EDUCATION

• University of Central Florida

Orlando, FL

Ph.D. of Civil Engineering (Transportation), GPA:4.0

Aug. 2022 – Dec.2024

• Chang'an University

Xi'an, China

M.E. of Traffic Engineering

Sep. 2020 – July.2023

• Chang'an University

Xi'an, China

B.E. of Traffic Engineering

Aug. 2016 – June.2020

WORK EXPERIENCE

• DiDi Research America

San Jose, CA

Safety System Engineer of Autonomous Driving

Jan. 2025 – Now

- Evaluated collision risk and conflict analysis for Level 4 autonomous systems.
- Developed scenario-based assessment practices and auto-labeling rules to advance safety case frameworks.
- Published research on modelling maneuver responses and collision severity in leading transportation journals.
- Led safety assessments with naturalistic driving data involving collisions and collision avoidance.
- Familiarity with STPA, ISO 21448, ISO 26262 and ISO 8800.

EXECUTIVE SUMMARY

- Experienced in collision avoidance modeling, assessing conflict scenarios and risks assessment for autonomous vehicles.
- Presented at **5+** international conferences and published **20+** papers **900+ citations**) in ranked journals, and served as reviewers for **20+** ranked journals, reviewed **50+** papers. (<https://scholar.google.com/citations?user=OZRGN3gAAAAJ&hl=en>).
- One paper on autonomous vehicle safety, covered by **200+** media outlets and ranked in the top **1%** on Altmetric.
- Performed statistical modeling and data analytics using **R, python, and SQL**.

MEDIA COVERAGE

Featured in Nature Communications Editor's Highlights (Engineering and Infrastructure section) for a paper on autonomous vehicle safety, covered by **200+** media outlets and ranked in the **top 1%** on Altmetric. Selected media reports include:

- Autonomous Vehicles Are Great at Driving Straight But humans still do better at turns and in the twilight
<https://spectrum.ieee.org/autonomous-vehicles-great-at-straight>
- Un estudio concluye que los coches sin conductor son más seguros que los llevados por personas
<https://elpais.com/tecnologia/2024-06-18/un-estudio-concluye-que-los-coches-sin-conductor-son-mas-seguros-que-los-llevados-por-personas.html>

HONORS

- 2025 TRB Highlighting Doctoral Student Research
- 2024 Editors' highlight paper (Top 1%) in Nature Communication
- 2021 ILUTM-7 CONFERENCE, Best Student Presentation
- 2023 UCF Doctoral Research Support Award
- 2023 UCF ORC Doctoral Fellowship
- 2022 UCF Graduate Presentation Fellowship Award

SELECTED PUBLICATIONS

*: Denote corresponding author

Autonomous Vehicles Safety Assessment:

- [1] **Ding, S.**, Abdel-Aty, M., Wang, D., Wang, Z., Zheng, O., Barbour, N., 2024. Exploratory Analysis of the Crash Severity between Vehicular Automation (SAE L2-5) with Multi-Source Data. **Accident Analysis & Prevention** Volume 206, 107692.
- [2] Abdel-Aty, M. and **Ding, S***, 2024. A matched case-control analysis of autonomous vs human-driven vehicle accidents. **Nature Communications**, 15(1), p.4931. (**Top 1% on Altmetric**)
- [3] **Ding, S.***, Abdel-Aty, M., Yang, S., & Wang, C. (2025). Assessing the safety effectiveness of advanced driver assistance systems. **Journal of Safety Research**, 95, 345-361.
- [4] Cui, Z., Dong, S., Sun, D. J., Ye, L., & **Ding, S.** (2024). How Does Environmental and Road Factors Impact Automated Vehicle (SAE Level 2) Crash Results? A Network and Coupling Analysis, accepted by Journal of Transportation Safety & Security.
- [5] **Ding, S.**, Abdel-Aty, M., Wang, D., Wang, Z., Zheng, O., Barbour, N., Crash Injury Severity Analysis under Different Levels of Driving Automation., Accepted at 104th Annual Meeting of the Transportation Research Board, Washington, D.C., 2025.
- [6] **Ding, S.**, Abdel-Aty, M., Wang, D., Wang, Z., Zheng, O., Barbour, N. AUTOMATED VEHICLE OPERATION INCIDENT DATASET AND EXPLORATORY ANALYSIS., Presented at 2024 Florida Automated Vehicle Summit.

Collision Risk Assessment:

- [1] Yuan, R., **Ding, S.**, Fang, Z., and Xiang, Q., 2023. Investigating the spatial heterogeneity of factors influencing speeding-related crash severities using correlated random parameter order models with heterogeneity-in-means. *Transportation Letters*, pp.1-13.
- [2] Rao, X., Yuan, C., **Ding, S.***, Fang, X., and Yuan, R. (2024). Analyzing Injury Severity of School Bus Crashes considering Temporal and Spatial Heterogeneity Using Random Parameters Ordered Probit Model., Accepted at 104th Annual Meeting of the Transportation Research Board, Washington, D.C., 2025.
- [3] Rao, X., Yuan, C., **Ding, S.***, Fang, X., & Yuan, R. (2024). Comparative Analysis of School Bus Crash Severity on Urban versus Rural Roadways: a random parameter logit model with mean heterogeneity., Accepted at 104th Annual Meeting of the Transportation Research Board, Washington, D.C., 2025.

Driving Behavior:

- [1] **Ding, S***, Abdel-Aty, M., Wang, Z. and Wang, D., 2024. Insights into vehicle conflicts based on traffic flow dynamics. *Scientific Reports*, 14(1), p.1536.
- [2] Yuan, R., **Ding, S.** and Wang, C., 2023. Comparative Analysis of Machine-Learning Models for Recognizing Lane- change Intention Using Vehicle Trajectory Data. *Infrastructures*, 8(11), p.156.
- [3] Wang, Z., Abdel-Aty, Zheng, O., M., **Ding, S.** Model Calibration for Microscopic Traffic Simulation Using Drone-Based Vehicle Trajectory Data, Presented at 103rd Annual Meeting of the Transportation Research Board, Washington, D.C., 2024.

Traffic System Approach:

- [1] **Ding, S.**, Yuan, C., Chen, J., Zhang, C. and Feng, J., 2023. Accessibility Optimization of Public Transportation in Historical Districts—A Study of Beilin District, Xi'an. In *CICTP 2023* (pp. 330-339).
- [2] Yuan, C., **Ding, S.**, Wang, Y., Feng, J. and Ma, N., 2023. Emergency Resource Layout with Multiple Objectives under Complex Disaster Scenarios, Presented at 103rd Annual Meeting of the Transportation Research Board, Washington, D.C., 2024.
- [3] Nie, J., Xia, S., Liu, Y., **Ding, S.**, Hu, L., Zhao, M., Fan, Y., Abdel-Aty, M., Preindl, M. and Jiang, X., 2023, June. A Data-Driven and Human-Centric EV Charging Recommendation System at City-Scale. In *Proceedings of the 14th ACM International Conference on Future Energy Systems* (pp. 427-438).
- [4] Zhang, C., Cheng, L., **Ding, S.** and Zhai, R., 2022. Analysis of Network Traffic Capacity Based on Travelers Considering Travel Behavior Choice. In *CICTP 2022* (pp. 2434-2444).
- [5] Yuan, C., Ma, N., Mao, X., Duan, Y., Zhao, J., **Ding, S.**, & Sun, L. (2024). Estimation of Greenhouse Gas Emissions of Taxis and the Nonlinear Influence of Built Environment Considering Spatiotemporal Heterogeneity. *Sustainability*, 16(16), 7040.

Large Language Model:

- [1] Huang, H., Zheng, O., Wang, D., Yin, J., Wang, Z., **Ding, S.**, Yin, H., Xu, C., Yang, R., Zheng, Q. and Shi, B., 2023. ChatGPT for shaping the future of dentistry: the potential of multi-modal large language model. *International Journal of Oral Science*, 15(1), p.29.
- [2] Zheng, O., Abdel-Aty, M., Wang, D., Wang, C. and **Ding, S.**, Trafficsafetygpt: Tuning a pre-trained large language model to a domain-specific expert in transportation safety, Presented at 103rd Annual Meeting of the Transportation Research Board, Washington, D.C., 2024